

SECTION A – CASE QUESTIONS (Total: 50 marks)

Answer **ALL** of the following questions. Marks will be awarded for logical argumentation/ calculation and appropriate presentation of the answers.

CASE

Future Intelligence Group ("the Group") has been engaged in artificial intelligence ("AI") and machine learning ("ML") research and development for two years. The Group is owned by two Doctor of Philosophy ("PhD") graduated from reputable universities who specialised in AI and ML. The Group has been using AI and ML to develop a generative AI model ("GAI") called FIChat that is able to produce new data based on patterns and structures learned from big data obtained in databases. It can produce content, such as texts, images, music, using deep learning methodologies and neural networks to analyse, understand and produce content that is similar to human-generated content. The Group has been providing FIChat services to registered and paid customers. The Group has signed a few AI, ML and GAI consultancy contracts to serve multinational companies. Due to the owners' academic backgrounds, the Group is specialised to serve students and researchers in universities in Hong Kong and this differentiates them from other GAI companies.

There are many companies using AI and ML to conduct the research and development of GAI in different countries. Some of them are supported by reputable multinational computer chipmakers and cloud computing companies with strong financial and technical support. This support is necessary due to the high capital requirements to develop GAI. Some companies are going out of business because they do not have sufficient capital to further research and develop their GAIs. Different companies have adopted different AI and ML methods to develop GAI that offers similar functions, but users are not aware of the technologies behind the GAI. Since there are different GAIs available in the market, customers have the option to choose and use GAIs that are suitable for their usages and requirements, and at a price within their budgets.

The development of GAI depends on fast and specialised computer chips called GPUs that are very expensive and are always in short supply because there are only a few computer chips manufacturers that produce these chips in developed countries. Additionally, the availability of GAI training professionals has been lagging behind compared to the rapid growth in research and development of AI and ML technologies such that companies must hire these professionals by offering high compensation packages and good career paths.

Aside from the high labor cost, costs to access databases used to train and develop GAIs, as well as the electricity and computer running costs, are very high and this makes research and development of GAI becoming not viable due to competitive GAI service fees charged to customers. Overall, the market of research and development of GAI has been dominated by a few large industrial players equipped with strong financial and technical capabilities. Small GAI companies like the Group have to find niche markets to provide specialised consultancy services to their customers in order to survive.

To increase revenue and profitability, the Group has been setting up a new consultancy division to offer AI, ML and GAI consultancy services to multinational companies and universities in Hong Kong from which it is growing a profitable business. The Group has set up different cost structures for different types of consultancy jobs. It is a challenge for the Group to hire AI, ML and GAI consultancy professionals in the industry because it needs to provide higher salary and innovative AI consultancy training and good career paths to consultancy professionals to serve customers in different projects.

To serve customers' specific needs and requirements, the Group allows customers to change specifications of service outputs within two weeks after signing the consultancy contract. The Group also plans to get back to all customers' enquiries within 24 hours. To improve customer service, the Group is considering setting up a specific performance measurement system to measure the performance of the AI, ML and GAI consultancy service business. The Group expects that the new performance measurement system should measure different dimensions of the consultancy service business and can provide a framework to assess the effectiveness of performance metrics. Moreover, the system can establish benchmarks to ensure the effectiveness of the performance measurement system. The Group has a vision to become the leading AI, ML and GAI consultancy service company in Asia.

The Group has been financed by the two owners' parents and friends and the Group will not consider to issuing more shares. Due to the high costs to develop FIChat and to pay GAI professionals, the capital is nearly used up and they require additional financing in the coming year. The Group has a good reputation in the AI, ML and GAI consultancy service industry and has won a few consultancy service prizes in Hong Kong. Therefore, the Group is planning to further develop their consultancy service business in Asian countries, and this will require additional capital.

The Group is in a breakeven financial situation in terms of profitability, but they are tight in terms of cash flow due to slow payments from multinational companies and universities. The revenue from other FIChat users is not sufficient to maintain and support further development of the Group's GAI business. Although the Group has positive working capital and fixed assets, such as computer servers, office equipment, to continue operation of the business, the Group needs to obtain additional financing to further develop FIChat and the new consultancy service business.

The Group does not have credit and borrowing histories. To obtain financing, one of the owners asked the financial controller, a certified public accountant, to meet his friend, a financial consultant, who agrees to obtain financing from a financial institute for the Group. During the meeting, the financial consultant suggested that the Group should provide an aggressive business plan and financial forecast based on his advice so that he can obtain financing for the Group. Moreover, the Group must pay him a 10% commission on the financing amount after he successfully obtains financing for the Group.

The financial consultant proposes two financing options to the Group. Option one is to issue 4,000 unsecured financial instruments at par with a face value of HK\$1,000 each. The instrument will pay 2.5% p.a. fixed coupon semi-annually in arrears and will mature in two years' time, redeemable at a premium of 4.5%. The comparable market-related interest rate on the instrument with similar risk is 6% p.a. Option two is to issue a redeemable convertible financial instrument with a total value of HK\$4 million and a face value of HK\$1,000 each. The instrument will pay a fixed annual coupon at 3.5% p.a. with a maturity of two years. The owners assume that each instrument can be converted to 50 ordinary shares of the Group. The Group's ordinary share price per book value is HK\$10 and the owners expect the book value of the share price to be HK\$20 based on their own valuations at the redemption date in two years. Similar convertible financial instruments are trading with a yield to maturity of 4% p.a. The tax rate is 16.5% and the above information is on a pre-tax basis.

Question 1 (15 marks – approximately 27 minutes)

- (a) Acting as the financial controller of the Group, prepare a memo to the owners of the Group that evaluates, with appropriate model, the industry attractiveness and profitability of the GAI industry.

Note: A maximum of 2 marks for communication skills will be awarded.

(10 marks)

- (b) Recommend, with justifications, whether the Group should further develop FIChat based on the evaluation in Question 1(a).

(5 marks)

Question 2 (10 marks – approximately 18 minutes)

Acting as the financial controller of the Group, recommend the most appropriate performance measurement model for the GAI consultancy service business and explain, with examples, how it can be used to achieve the Group's performance measurement objectives.

(10 marks)

Question 3 (15 marks – approximately 27 minutes)

Based on the existing financing information and requirements of the Group,

- (a) Besides the TWO financing options proposed by the financial consultant, recommend and justify, without any calculation, suitable and feasible financing options that the Group shall pursue to obtain additional financing for supporting the business operations and development of the Group.

(10 marks)

- (b) Discuss ethical principles that the financial controller should consider in obtaining financing based on the advice of the financial consultant. Advise the financial controller on how to handle the ethical issues.

(5 marks)

Question 4 (10 marks – approximately 18 minutes)

Based on the advice of the financial consultant, calculate the current fair value of each of the financial instruments and recommend a financing instrument to the Group.

Note: A maximum of 2 marks for analytical skills will be awarded.

(10 marks)

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End of Section A

SECTION B – ESSAY / SHORT QUESTIONS (Total: 50 marks)

Answer **ALL** of the following questions. Marks will be awarded for logical argumentation/ calculation and appropriate presentation of the answers.

Question 5 (17 marks – approximately 31 minutes)

A retail company, Tiger Watches ("Tiger"), sells luxury watches and is considering purchasing the property from which they are renting for their watch shop, so that they can save rent, which is their key operating cost. After negotiation, Tiger received an offer from the landlord to purchase the property at HK\$240 million. Currently, Tiger is paying HK\$1 million in rent per month to the landlord for this premises and expects that there would be 10% rental increase every year based on Tiger's turnover increase.

Tiger estimates that after successfully purchasing this property, the transaction cost, including agency fee, legal fee, and stamp duty, would be HK\$31.2 million, and they would spend HK\$10 million for renovation. Tiger considers using the property for five years. Based on the advice from the surveyor, assuming that there is no major fluctuation in the property market, Tiger could sell the property at the same consideration in five years at HK\$240 million while the buyer at that time would be responsible for most of the transaction cost.

The cost of capital of Tiger is 4%.

Required:

Ignore the accounting treatment of HKFRS 16 in answering Question 5(a), (b) and (c) below.

- (a) Assuming that the renovation period is immaterial to the whole project, calculate the net present value for the purchase of this property. Recommend with justification if Tiger should accept this offer.
(8 marks)
- (b) Advise Tiger, without any calculation, the possible factors under the control of Tiger that could be changed to reach a different recommendation from Question 5(a) above.
(3 marks)
- (c) Advise Tiger any other qualitative factors that should be considered for the purchase.
(6 marks)

Question 6 (14 marks – approximately 25 minutes)

Superb Café ("Superb") is a chain coffee shop based in Hong Kong and listed on the main board of the Hong Kong Stock Exchange ("HKEX"). Superb currently has 25 coffee chain stores in Hong Kong. After the pandemic, Superb is expecting a growth in demand for catering services, including coffee shops. They are considering expanding their channel by acquiring another coffee chain store, Target Coffee ("Target"), which is a private company with 30 stores in Hong Kong and Macau.

While negotiating, Target agreed with Superb to conduct due diligence for the acquisition.

Required:

- (a) Advise Superb on the due diligence process from the operation perspective ("operation due diligence") regarding the review and verification for the acquisition.
(6 marks)
- (b) Besides operation due diligence, without going into details, identify other due diligence processes that should be considered. Recommend Superb on who could support those due diligence processes and explain their roles.
(5 marks)
- (c) Advise Superb, as a listed company on the main board of the HKEX, the ESG discussions that must be included in the Business Review Section of the Director's Report.
(3 marks)

Question 7 (19 marks – approximately 34 minutes)

Happy Purple Limited ("Purple") is a toy design and manufacturer who sells their toys to major department stores in Europe. The management office of Purple is based in Hong Kong and they also operate a design workshop in Hong Kong. Purple operates a few factories and sources raw materials in mainland China. They pay their raw materials vendors and the overheads of their factories in RMB, while they pay the expense of the Hong Kong management office and design workshop in HKD. They receive EUR for their sales to their European customers.

Purple had just fulfilled a significant sales order to Europe for a total of EUR10,000,000 and their credit term is 90 days, so Purple is expected to receive payment for this sales order after 90 days. Purple noted that EUR had been depreciating since the beginning of the year and the bank forecast that the exchange rate of EUR to HKD would become 1:8 after three months. Purple worries that EUR would further depreciate so they are considering hedging half (50%) of their position.

The exchange rate of EUR to HKD stands at 1:8.5 today. Purple got a bank quote for a 3-month forward contract for EUR to HKD at a rate of 1:8.35.

Before making the hedging decision, Purple referred to advance analytics and summarized the probabilities for following scenarios.

Scenarios	Probability
The bank's forecast of EUR to HKD after three months is accurate	65%
The exchange rate of EUR to HKD after three months stays the same as today	25%
The exchange rate of EUR to HKD become 1:9 after three months	10%

Required:

- (a) According to the above scenarios, calculate the expected net exchange gain/ loss in HKD with and without hedging. Recommend Purple on whether to hedge or not.
(7 marks)
- (b) Purple explored the alternative of entering a call option, where they would sell EUR and buy HKD at a rate of 1:8.35 after THREE months by paying a 3% premium for the option. Advise Purple, with calculation, on whether this alternative should be considered instead of the forward contract.
(6 marks)
- (c) Compare the advantages and disadvantages of using forward and option to hedge as a hedging tool.
(6 marks)

* * * END OF EXAMINATION PAPER * * *